

# SHENG ZHAO

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*My research interests lie in **ubiquitous and social computing in virtual reality**, as well as **3D generation by diffusion model**.*

## EDUCATION

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**Tsinghua University, Beijing, China** 2021.09 – Present  
B.Sc. in **Mathematics and Physics** & B.Eng. in **Electrical Engineering** Dual degree  
Junior, GPA: 3.81/4.

## PUBLICATIONS & MANUSCRIPTS

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- **Sheng Zhao**, JunRay Zhu, Xueyang Wang, Hongyi Li, Fang Yi, Shuning Zhang and Xin Yi<sup>†</sup>. CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination. *UbiComp'24*, under review. [PDF](#).
- Xueyang Wang\*, **Sheng Zhao\***, Yihe Wang, Ziyu Han, Xin Tong<sup>†</sup> and Xin Yi<sup>†</sup>. EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality. *ISMAR'24*, in submission.

## RESEARCH EXPERIENCES

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HCI Group, Tsinghua University, China 2023.04 – 2024.01

- *Research Assistant, Advisor:* [Xin Yi](#), [Yuanchun Shi](#)
- **EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality**
  - Implementing VR face2face chat using TCP-based socket and [Oculus SDK](#).
  - Designing both qualitative and quantitative research methods based on interviews and questionnaires, conducting user studies and data analysis.
  - Proposing an improved camera2blendshape method to mitigate the uncanny valley effect and enhance the emotional experience.

We'll submit it to [ISMAR'24](#). Detailed description and code are [here](#).

HCI Group, Tsinghua University, China 2023.10 – 2024.01

- *Research Assistant, Advisor:* [Xin Yi](#)
- **CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination**
  - Project leader, proposing the current two-factor authentication framework based on unlock pattern and behavior biometrics.
  - Using Unity to build the 3x3 unlock UI leveraging gaze interaction in VR.
  - Designing the biometric classifier by machine learning, using RF, etc. to implement the classification of behavior biometrics during saccades and fixations of eye movements.
  - Conducting user studies, validating framework robustness against password collision, contextual factors, and shoulder surfing attacks.

We have submitted it to [UbiComp'24](#). Download [PDF](#). Detailed description are [here](#).

**Ongoing** research experiences can be found [here](#).

## TECHNOLOGY COMPETITIONS

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**Tsinghua University Software Design Competition** 2022.10 – 2023.01

Developed a Smart Home Android app using Java, which includes features like bluetooth connectivity, weather forecast queries and communication with hardware devices. I won the Second Prize. Code [here](#).

## The 12th Tsinghua University Digital System Design Competition

2022.08 – 2022.10

Designed a small car driven by MSP430 with various sensors for autonomous navigation and resource acquisition. I won the First Prize(1/60). Details and Videos are [here](#).

## SKILLS

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- Program languages: C/C++/C#, Python, Java, MATLAB and Verilog.
- Tools and Frameworks: Git, LaTeX, Unity, PyTorch, Scikit-learn, Markdown and MSP430.
- Techniques: VR development, Quantitative and Qualitative research.

## SELECTED AWARDS

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| <i>First Prize</i> , 36th China Physics Olympiad (CPHO), 0.9%                                | 2020       |
| <i>First Prize</i> , The 12th Tsinghua University Digital System Design Competition, 1.7%    | 2022       |
| <i>First Prize</i> , Tsinghua University Scientific And Technological Innovation Scholarship | 2023       |
| <i>First Prize</i> , Tsinghua University Sports Excellence Scholarship                       | 2022, 2023 |
| <i>First Prize</i> , The 3rd Tsinghua Craftsman Competition, 1%, Presented a speech          | 2023       |